

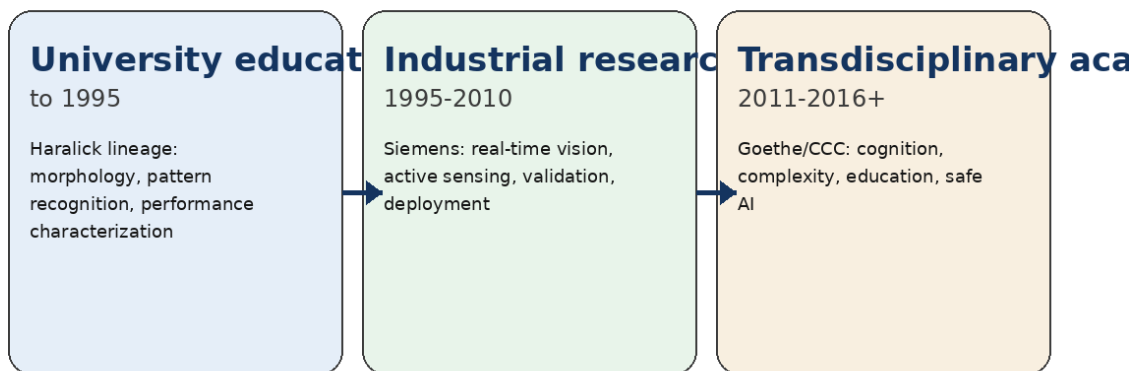
Counting with Uncertainty

A Transdisciplinary Monograph on Computer Vision, Artificial Intelligence,
Systems Engineering, and a Personal Research Journey

Draft prepared for Visvanathan Ramesh

University education and research; industrial scaling; transdisciplinary integration in
academia

Counting with uncertainty: from vision algorithms to sy



Unifying framework: Context + Tasks + Performance requirements -> Hardware

This monograph is a structured draft for revision, teaching, and webpage development. It synthesizes uploaded primary materials, historical background, and current AI context.

Preface: Why This Monograph Exists

This monograph was written to make a journey visible. It is not meant to be a conventional autobiography, a technical survey, or a personal webpage in long form, although it contains elements of all three. It is meant to be a transdisciplinary map: a way for a young student to see how a life in computer vision and artificial intelligence can grow through many intellectual habitats - electrical engineering, pattern recognition, image processing, artificial intelligence, applied mathematics, philosophy, psychology, neuroscience, industrial systems engineering, and teaching.

The immediate historical object behind this monograph is the 2008 CVPR panel presentation on future directions in computer vision and pattern recognition. That presentation looked backward to the 1960s and 1970s, described the modular vision systems of the 1980s, the statistical and contextual turn of the 1990s, and the systems-engineering concerns of the 2000s. It also looked forward to active sensing, multimodal fusion, automated reasoning, reflexive and reflective vision, and the emergence of a vision and audio systems science[1]. From the standpoint of 2026, the presentation reads not simply as a prediction about computer vision but as a compressed philosophy of intelligent systems.

The deeper purpose of this monograph is to stratify the author's journey in three layers. The first layer is university education and research up to the 1995 dissertation, under the mentorship of Robert M. Haralick at the University of Washington. The second layer is industrial research at Siemens Corporate Research, where the ideas of performance characterization, statistical modeling, and system validation were scaled to real-time and real-world applications until roughly 2010. The third layer is the academic and educational integration from 2011 onward at Goethe University Frankfurt, including the move toward transdisciplinary computing education, cognitive systems, neurotechnology, systems engineering for intelligence, and safe human-AI co-evolution[8]. The period 2011-2016 is treated as a distinct transition because it marks the emergence of a new academic identity: no longer only computer vision or industrial AI, but systems engineering for intelligence.

The monograph is written for a transdisciplinary audience. A mathematically trained reader should see why philosophy and psychology matter. A humanist should see why edge detection, morphology, and performance characterization are not merely technical details. A student should see that an intellectual journey is not a straight line but a succession of reframings. A systems engineer should see that AI safety and AI governance are not late additions to artificial intelligence,

but natural consequences of deploying perception and reasoning in the world.

Reading note. This document is a draft artifact. It is deliberately self-contained, occasionally reflective, and pedagogical. It uses the first person in a few places to support eventual adaptation into a personal webpage. Every such passage should be reviewed by the author before publication.

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Chapter 1

The Journey at a Glance

1.1 Three Strata of a Blessed Journey

A personal research journey can be narrated as a list of positions, publications, and grants. Such a list is useful, but it hides the inner structure of the journey. The more illuminating structure is epistemic: what counted as knowledge at each stage? What kinds of systems were imagined? What forms of evidence mattered? What risks became visible? What kinds of students and collaborators could be formed by the work?

The journey reconstructed here has three major strata.

Table 1.1: Three strata of the journey

Period	Dominant setting	Central transformation
University education and research, to 1995	University of Washington; Robert Haralick's group; DARPA Image Understanding context	From image algorithms to performance characterization: vision components are not black boxes but estimators whose errors can be modeled, propagated, and optimized.
Industrial research and scaling, 1995-2010	Siemens Corporate Research; real-time vision and modeling; security, mobility, healthcare, energy	From algorithm evaluation to real-world systems engineering: context, tasks, and performance requirements drive sensor choice, module selection, parameter tuning, validation, and deployment.
Academic transdisciplinary integration, 2011-2016 and beyond	Goethe University Frankfurt; neurotechnology, cognition, computation, education	From vision systems to systems engineering for intelligence: integrate computation, cognition, complexity, psychology, neuroscience, philosophy of AI, and pedagogy.

The three strata do not replace one another. The dissertation-era insight that an algorithm’s output perturbations must be related to its input perturbations remains alive in industrial vision systems, in active sensing, in autonomous driving, and in contemporary AI governance. Similarly, the industrial insight that a system must be designed from user requirements, world context, task structure, and performance constraints becomes the basis for an educational framework in which students learn to think from context to system architecture. In the 2025 curriculum article, this becomes the framework $(C, T, P) \rightarrow \text{HW/SW Sys}$ [8].

1.2 The Core Motif: Counting with Uncertainty

The German 2018 article, whose headline asks how reliable artificial intelligence is, uses the phrase *Mit Unsicherheiten rechnen*: computing, or reckoning, with uncertainties[7]. That phrase captures the continuity of the journey. The image-understanding dissertation is about uncertainty in features, edges, lines, corners, and algorithm sequences[2]. The Siemens work is about uncertainty in real-time surveillance and automotive systems[4]. The DARPA performance-modeling summary is about uncertainty in sensors, platforms, targets, environments, algorithm choices, tuning parameters, and system-of-systems operation[5]. The later work on certifiable autonomous cognitive systems is about uncertainty in world models, cyber-physical gaps, decision-making, and certification[6]. The educational work is about preparing students to reason with uncertainty across technology and society[8].

The recurring question is therefore not merely, “How can a machine see?” It is:

How can a designed system act intelligently when its measurements are imperfect, its world is changing, its tasks are contextual, its models are partial, and its consequences matter?

1.3 A First Map of the Intellectual Lineage

The journey can also be expressed as a lineage of questions:

1. **Pattern recognition:** How can measured patterns be classified, clustered, represented, and discriminated?
2. **Image processing:** How can images be filtered, segmented, enhanced, and transformed into meaningful structures?
3. **Mathematical morphology:** How can spatial structure be manipulated through operators with formal properties?
4. **Symbolic AI:** How can knowledge, rules, and proofs generate procedures and explanations?

5. **Performance characterization:** How do input perturbations propagate to output perturbations through algorithm components?
6. **Systems engineering:** How do context, task, performance, architecture, implementation, and validation cohere?
7. **Cognitive systems:** How can fast reflexive processes, slower reflective reasoning, world knowledge, and control loops be combined?
8. **Responsible AI:** How can systems be made reliable, explainable, certifiable, auditable, and aligned with human flourishing?

The rest of the monograph expands this map.

Chapter 2

Origins: Vision, AI, Pattern, and Mind in the 1960s and 1970s

2.1 Computer Vision Did Not Begin as One Field

Computer vision is often narrated as a single technical field that progressed from hand-crafted features to machine learning and then to deep learning. This is historically convenient but intellectually misleading. The early field was a confluence of several streams: statistical and syntactic pattern recognition, digital image processing, symbolic artificial intelligence, signal processing, applied mathematical pattern theory, psychophysics, neuroscience, cybernetics, and philosophy of mind.

The 2008 CVPR presentation compresses the 1960s and 1970s into three streams: statistical/syntactic pattern recognition, early image processing, and parallel efforts in artificial intelligence using symbolic descriptions for objects[1]. This is an excellent start, but the transdisciplinary view requires more layers.

2.2 AI as an Independent Thread

Artificial intelligence emerged with ambitions much broader than computer vision. The Dartmouth proposal of 1955, circulated by McCarthy, Minsky, Rochester, and Shannon, framed intelligence in terms of language, abstraction, problem solving, self-improvement, and machine simulation of human-like abilities[9]. Early AI included theorem proving, chess, symbolic logic, planning, natural language, knowledge representation, and robotics. Newell and Simon's physical symbol system hypothesis framed intelligence as symbol manipulation and heuristic search[12]. Robinson's resolution principle gave automated theorem proving a powerful inference method[11]. McCarthy's work on common sense treated the representation of knowledge and inference as a central problem[10].

This AI thread matters for computer vision because it defined a goal that low-level image processing alone could not satisfy: a machine should not merely detect edges or classify patterns;

it should produce descriptions that can support reasoning, planning, explanation, and action. Blocks-world systems, Shakey the Robot, and early natural-language systems like SHRDLU were simplified worlds, but they forced the question of grounding: how does perception connect to symbols, and how do symbols connect to action?

2.3 Pattern Recognition and the Measurement View

In statistical pattern recognition, the image is a source of measurements. It can be transformed into features, and those features can be classified under statistical assumptions. This tradition is strongly associated with decision theory, discriminant functions, clustering, nonparametric methods, and later Bayesian methods. Duda and Hart's *Pattern Classification and Scene Analysis* helped codify this view for a generation of researchers[13].

Robert Haralick's work belongs centrally to this lineage. His texture features, based on gray-tone spatial dependence, showed that perceptual qualities such as texture can be made computable by modeling spatial relationships among gray tones[14]. This was not just a technical advance; it was an epistemic move. It said that visual regularity is not merely something humans see; it can be represented as a statistical relation among measurements. That lesson persists in modern learned representations, even when the features are no longer hand-designed.

2.4 Digital Image Processing and the Rosenfeld Tradition

Azriel Rosenfeld's work helped create digital image processing and image analysis as coherent research domains. The image-processing tradition treated images as signals and arrays that could be filtered, segmented, labeled, transformed, and measured. It gave computer vision a language of edges, regions, boundaries, connected components, skeletons, contours, and textures. Many later debates about recognition and reasoning presupposed this lower-level infrastructure.

This tradition should not be dismissed as pre-AI. In practice, every AI system that sees must ultimately depend on measurement, resolution, noise, quantization, sampling, sensor geometry, and transformation. In 2026, even vision-language foundation models depend on image formation, data curation, sensor modalities, annotation practices, compression artifacts, and preprocessing pipelines. The signal never disappears; it is hidden inside the training system.

2.5 Applied Mathematics and Grenander's Pattern Theory

Ulf Grenander's pattern theory provides a deeper mathematical thread. In ordinary pattern recognition, a pattern may be a point in feature space. In Grenander's pattern theory, a pattern is a structured configuration generated by parts, relations, transformations, and probability laws[15]. Pattern theory therefore asks: what is the space of possible patterns? What transformations

preserve identity? How do complex patterns arise from simpler generators? How can we synthesize patterns, not merely analyze them?

This is one of the most important bridges to the 2026 context. Generative models, diffusion models, neural fields, differentiable rendering, procedural world generation, and simulation-based evaluation all revive the older question of pattern synthesis. The form has changed, but the question remains: how do lawful structures generate the variability we observe?

2.6 Signal Processing, Cybernetics, and Control

Signal processing contributed a different vocabulary: noise, filtering, sampling, estimation, bandwidth, correlation, detection, and reconstruction. Shannon’s information theory formalized communication under noise[16]. Wiener’s cybernetics framed organisms and machines in terms of feedback and control[17]. Kalman filtering made recursive state estimation a foundational engineering method[18]. Gabor’s work on time-frequency localization anticipated later filter-bank and wavelet ideas[19].

These ideas matter because a vision system deployed in the world is not a static classifier. It senses, estimates, updates, acts, and receives new evidence. The active-sensing loop in later performance-modeling work - acquire, process, estimate state, choose next sensor action - is a descendant of this control tradition[5].

2.7 Psychology, Philosophy, and the Meaning of Perception

Psychology and philosophy provide the deepest questions. Helmholtz’s idea of unconscious inference treats perception as an inferential construction. Gestalt psychology treats perception as organization into meaningful wholes. Gibson’s ecological psychology treats perception as the pickup of information relevant to action and affordance. Marr’s computational theory of vision asked what problem the visual system solves, what representations and algorithms solve it, and how those algorithms are physically implemented[20].

Perceptual Control Theory, developed by William T. Powers and collaborators beginning in the late 1950s and published in book form in *Behavior: The Control of Perception* in 1973, adds a crucial inversion: behavior is not the controlled output; perception is what the organism controls[21]. This idea is essential for a transdisciplinary theory of vision systems. A visual system is not merely a representation machine; it can be part of a closed loop that acts to maintain, refine, or control perceptual variables.

Aaron Sloman’s philosophy of AI adds another important layer. Sloman asked not only whether machines can think, but what varieties of mind and architecture are possible[22]. His idea of “productive laziness” - intelligent systems avoid unnecessary work by using appropriate abstractions, short cuts, and layered architectures - resonates with real-time vision systems that use priors for fast hypothesis generation and reserve detailed reasoning for selected cases. This will reappear in

the discussion of reflexive and reflective vision.

Chapter 3

The 1980s: Modular Vision and the Dream of the Pipeline

3.1 The Canonical Pipeline

By the 1980s, computer vision often imagined a system as a pipeline: low-level feature extraction, mid-level grouping, and high-level matching or interpretation. The 2008 CVPR deck describes this clearly: edges, lines, corners, and regions at the low level; perceptual grouping at the middle level; and matching of feature groupings to models at the high level[1].

This pipeline had great explanatory value. It made it possible to divide labor. One researcher could study edge detection, another grouping, another geometric matching, another CAD-based recognition. It also aligned loosely with psychological ideas about early, middle, and high-level vision. But the pipeline hid several hard problems.

3.2 The Hidden Problems of the Pipeline

First, errors propagate. A wrong edge threshold can destroy a line segment; a broken line segment can destroy a junction; a missing junction can destroy model matching. Second, each stage changes the data representation. Pixels become edgels, edgels become chains, chains become lines or arcs, lines become relational structures, and relational structures become model hypotheses. Third, parameter choices are not neutral. A threshold that works in one lighting condition can fail in another. Fourth, the appropriate criterion for a component depends on the final task. A boundary extractor for inspection, road following, medical segmentation, and surveillance may require different tradeoffs.

These are precisely the problems that later become performance characterization and systems engineering. The 1980s gave the field modular structure; the 1990s had to ask how such modules could be characterized, composed, tuned, and validated.

3.3 Model-Based Vision and Geometry

The 1980s also saw major work in model-based recognition, shape representation, geometric invariants, motion, optical flow, shape from shading, shape from texture, and segmentation as optimization. Thomas Binford’s generalized cylinders and model-based vision helped frame objects as structured 3D entities rather than flat label categories[23]. Horn’s shape-from-shading work exemplified physics-based vision: images are generated by physical interactions among surfaces, illumination, reflectance, and sensors[24].

The geometric and physics-based tradition was not defeated by data-driven learning. It was submerged for a time under benchmarks and deep recognition, but it has returned in 3D vision, neural rendering, robotics, autonomous driving, embodied AI, and world modeling. In a deployed autonomous system, geometry is not optional.

Chapter 4

The Haralick School and the University of Washington Context

4.1 A Mentor and an Intellectual Culture

The author’s graduate trajectory begins in 1987 at the University of Washington under Robert M. Haralick. This matters not only biographically but epistemically. Haralick’s research culture emphasized mathematical precision, measurement, statistical characterization, morphology, pattern analysis, and the disciplined formulation of vision problems. It was a culture in which an image-processing operator was not just a piece of code; it was an object to be defined, analyzed, evaluated, and understood.

In this context, the student’s question changed from “What algorithm works?” to “What does it mean for an algorithm to work, under what assumptions, with what perturbations, and for what task?”

4.2 Automatic Morphology: Hyonam Joo’s Dissertation

Hyonam Joo’s 1991 dissertation, *Automatic Morphology*, is an important companion work. It developed a system that accepts a vision task specification and uses knowledge of mathematical morphology to construct a procedure that executes the task[3]. The dissertation represents the states of images and their relationships in predicate calculus, translates English descriptions of morphological procedures into predicate logic, and uses search and proof to generate morphological processing sequences. Each proof step corresponds to a morphological operation; the proof becomes the program.

This is a remarkable fusion of mathematical morphology and symbolic AI. It is not merely “image processing.” It is automatic programming for vision based on formal operator knowledge. It belongs to the tradition of theorem proving, knowledge representation, and Sloman-style design-space exploration, but grounded in concrete visual operations.

4.3 The Complementarity of Joo and Ramesh

Joo's question was: given a task and a knowledge base of morphological operators, can a system synthesize an appropriate procedure? The author's dissertation question was complementary: once a vision algorithm sequence is selected, can its performance be characterized, its parameters selected, and its behavior predicted over an application domain? Together they form a duality.

Table 4.1: Two complementary dissertations in the Haralick lineage

Dimension	Joo: Automatic Morphology	Ramesh: Performance Characterization
Main problem	Construct morphological procedures from formal task specifications.	Characterize and optimize algorithm sequences for image-understanding tasks.
Knowledge form	Predicate calculus, theorems, operator knowledge, proof search.	Probabilistic perturbation models, application-domain priors, criterion functions.
AI lineage	Symbolic AI, theorem proving, automatic programming.	Statistical decision theory, system identification, experimental design.
Vision lineage	Mathematical morphology and binary image operators.	Feature extraction, grouping, matching, edge/line/corner analysis.
Deep question	Can visual procedures be synthesized from declarative knowledge?	Can visual systems know and optimize their own reliability?

This duality foreshadows the later systems-engineering vision: intelligent systems require both procedure synthesis and performance self-knowledge.

Chapter 5

The Dissertation: Performance Characterization of Image Understanding Algorithms

5.1 The Dissertation Problem

The 1995 dissertation, *Performance Characterization of Image Understanding Algorithms*, states the problem with unusual clarity. Image Understanding systems are complex. They are composed of algorithm sequences. A model-based recognition system has feature extraction, grouping, and model matching. Each component often has tuning parameters. Traditionally these parameters were chosen by trial and error or empirical data. The dissertation proposes a methodology for analysis and design following systems-engineering principles[2].

The key definition is performance characterization: establishing the correspondence between random perturbations in the input and random perturbations in the output. This is system identification for vision algorithms. Given an ideal input model, an input perturbation model, an output perturbation model, and a criterion function, the performance of an algorithm can be characterized as a function of tuning parameters. With application-domain annotation, prior distributions over ideal input parameters can be estimated, validated, and used to tune parameters over an input population[2].

5.2 Why This Was a Systems-Engineering Move

The dissertation does not merely evaluate algorithms. It reframes vision algorithms as components in engineered systems. The central systems questions are:

1. What is the input population?
2. What is the ideal input model?

3. What perturbations corrupt the input?
4. How does an algorithm transform those perturbations?
5. How does the representation change from pixels to edgels, chains, line segments, corners, or matches?
6. What criterion function matters for the final task?
7. How should parameters be selected to satisfy performance requirements?

This is why the dissertation is not only an image-processing contribution. It is an early articulation of AI systems engineering.

5.3 Changing Data Units

One of the dissertation’s most important observations is that vision algorithms change the data unit. An edge detector maps pixels to edgels. A linker maps edgels to chains. A segmentation or extraction algorithm maps chains to curve segments. A model matcher maps segments to hypotheses. Therefore, the error representation must also change. Pixel-level false alarms and misdetections are not enough to characterize line extraction; line extraction requires fitting errors, segmentation errors, and perturbations in fitted coefficients. This insight remains highly relevant. In a modern deep system, pixels become tokens, tokens become embeddings, embeddings become attention states, attention states become detections or captions, and outputs become actions. Each representation change can hide or amplify uncertainty.

5.4 Application Domain Characterization

The dissertation’s use of annotation to characterize application-domain priors is a major forerunner of later domain modeling. It anticipates the 2008 CVPR claim that the key to rapid commercialization lies in systematic specification, translation of specification to design, and validation of designed systems[1]. It also anticipates modern concerns with data curation, benchmark design, simulation, scenario coverage, and domain shift.

5.5 The Deeper Lesson

The dissertation’s deepest lesson is epistemic humility. A vision system should not only output an answer; it should know the conditions under which the answer is reliable. This idea will later become performance modeling, self-diagnosis, active sensing, certifiability, and responsible AI.

Chapter 6

Industrial Research: Scaling to Real-World Vision Systems

6.1 From Dissertation to Siemens Corporate Research

The move from university research to Siemens Corporate Research transformed the problem scale. In academia, the dissertation could focus on algorithm sequences and their theoretical and empirical characterization. In industry, the question became: how can real systems be designed, tuned, validated, deployed, maintained, adapted, and trusted under customer requirements and real operating conditions?

The application domains expanded: security, surveillance, traffic control, automotive perception, healthcare, industrial inspection, energy, and monitoring. The algorithms expanded: detection, tracking, registration, segmentation, recognition, sensor fusion, Bayesian networks, context models, and optimization. The organizational problem also expanded: large teams, product constraints, customer requirements, deadlines, and integration.

6.2 Greiffenhagen’s Dual-Camera Surveillance System

Michael Greiffenhagen’s dissertation, supervised locally by the author at Siemens Corporate Research, is a landmark example of the industrial scaling step. The system combines an omnidirectional camera for wide-area detection with an active pan-tilt-zoom camera that provides a high-resolution zoomed image of a person’s head. The system’s parameters and their uncertainties depend on geometry, lighting, contrast, relative position, calibration error, and sensor noise. Crucially, the uncertainty estimate is used to set the zoom factor so that, with a user-specified probability, the head is contained in the active camera image[4].

This is performance characterization as action. The uncertainty model is not merely diagnostic. It controls the sensor. The system is therefore not just a passive recognizer; it is an active perceptual system. It senses, estimates, controls, and validates.

6.3 Reuse Through Systematic Design

A second contribution of Greiffenhagen’s work is reuse. A system designed for indoor static illumination was extended to quasi-outdoor dynamic illumination with minimal redesign because the original design used suitable modules and statistical representations[4]. This is a powerful industrial lesson: the value of systems engineering is not only better initial performance but extensibility, modularity, diagnosis, and reuse.

6.4 The Siemens Systems-Engineering View

The 2008 CVPR slides describe a systems design and validation process: system requirements, application priors, system design, system configuration, analysis and optimization, performance characteristics, performance characterization, and automatic programming[1]. This is essentially the industrialized version of the dissertation program.

The workflow is:

1. Specify application constraints, tasks, and performance tradeoffs.
2. Translate the specification into a probabilistic context representation.
3. Use the representation to identify filters or operators appropriate to the task.
4. Fuse filter responses using a statistical scheme designed for the context.
5. Validate statistical models and system performance across tuning parameters.
6. Optimize computation-accuracy tradeoffs.

This is a bridge from computer vision to systems science.

Chapter 7

Performance Modeling, Active Vision, and Systems of Systems

7.1 DARPA Performance Modeling Summary

The 2013 DARPA performance-modeling summary consolidates the performance-characterization doctrine. It distinguishes analytical, white-box modeling from empirical, black-box evaluation. In the white-box approach, each component is treated as an estimator whose deterministic and stochastic characteristics are derived as a function of inputs; the total system is a compound estimator. In black-box evaluation, end-to-end performance is measured as a function of inputs, design choices, and tuning parameters through careful experimental design[5].

The document also emphasizes why black-box testing is insufficient for active sensing systems and systems of systems: the combinatorial explosion of sensors, targets, platforms, environments, algorithms, and conditional interactions can make exhaustive testing prohibitive[5].

7.2 Active Vision as a Control Loop

The DARPA summary describes active vision as a loop: acquire or sense, process or interpret, estimate the state of understanding, choose sensor parameters or actions for the next step, and repeat until a global task objective is achieved[5]. This view has deep connections to cybernetics, Perceptual Control Theory, and modern robotics. It also connects to the 2008 reflexive/reflective architecture.

Active vision is not merely vision with a moving camera. It is vision with a performance model. The system chooses the next measurement based on expected information gain, uncertainty, task relevance, and cost. This is productive laziness in action: do not compute or sense everything; sense what matters for the task.

7.3 From Performance Evaluation to Performance Prediction

Performance evaluation asks how a system did on a dataset. Performance prediction asks how the system is likely to perform under a scenario, before or during deployment. Prediction enables adaptation, reconfiguration, sensor control, parameter tuning, and runtime diagnostics. This is the shift from post-hoc evaluation to anticipatory systems intelligence.

Chapter 8

The 2008 CVPR Panel as a Historical Hinge

8.1 Why the 2008 Presentation Matters

The 2008 CVPR panel presentation is a compact statement of an intellectual program. It begins with the history of vision technology, moves through statistical learning and systems engineering, gives concrete examples, and ends with future topics: multimodal fusion, grammars, cognition, situational understanding, scalable search, HMI integration, active control, and interdisciplinary teams[1].

The slides are interesting in retrospect because they do not merely predict the rise of better algorithms. They predict the need for a systems science of vision and audio, one that integrates AI, statistical learning, neuroscience, cognitive science, psychology, computer vision, and engineering.

8.2 The Reflexive and Reflective Architecture

One of the most philosophically rich slides is the real-time vision system design principle of “human-like reasoning: reflexive and reflective reasoning”[1]. It separates a reflexive vision component for real-time hypothesis generation from reflective reasoning for detailed estimation and belief revision. It includes world knowledge, normal pattern statistics, 3D scene models, priors, rules, sensor parameter selection, and a tradeoff between computational speed and accuracy.

This architecture can be read in several ways:

- As an engineering architecture for real-time vision.
- As a cognitive architecture with fast and slow processes.
- As a Sloman-like layered architecture with reactive and deliberative levels.
- As a PCT-compatible loop in which sensing and action control perceptual variables.

- As an early version of what modern AI calls tool routing, model routing, test-time computation, and adaptive inference.

8.3 The Future Topics Revisited from 2026

The presentation’s future topics can be mapped to current AI developments.

Table 8.1: CVPR08 future topics viewed from the 2026 context

CVPR08 future topic	2026 retrospective interpretation
Multimodal fusion	Vision-language models, audiovisual perception, radar/lidar-camera fusion, embodied agents, medical multimodal systems.
Pattern models and grammars	Scene graphs, neuro-symbolic AI, program synthesis, compositional generalization, structured prompting, event grammars.
Cognition and situational understanding	Video-language reasoning, autonomous driving, robotics, activity understanding, world models, and context-aware decision systems.
Advanced systems engineering	AI risk management, model cards, dataset documentation, runtime monitoring, simulation-based validation, certification.
Scalable search and mining	Large-scale retrieval, vector databases, multi-modal search, foundation model indexing, medical case retrieval.
Better HMI and automated analytics	Human-AI teaming, explainable AI, cognitive engineering, interactive copilots, educational AI systems.
Active and real-time control	Embodied AI, active perception, closed-loop autonomy, adaptive sensing, test-time planning and control.

The striking point is that the 2008 questions did not vanish with deep learning. They became more urgent.

Chapter 9

The Deep Learning Era and Its Reframing of the Old Questions

9.1 What Deep Learning Changed

The deep learning era changed the center of gravity of computer vision. Large datasets, GPUs, convolutional networks, transformers, contrastive learning, self-supervised learning, diffusion models, and foundation models made representation learning dominant. ImageNet-scale supervised learning and later vision transformers and vision-language models showed that large learned systems could outperform hand-crafted pipelines on many benchmarks[25, 26, 28, 27].

From one perspective, this displaced the old modular pipeline. From another perspective, it internalized the pipeline inside large learned architectures. Feature extraction, grouping, and semantic interpretation still occur, but not as hand-labeled modules. They occur as learned intermediate representations.

9.2 What Deep Learning Did Not Solve

Deep learning did not eliminate the need for context, task specification, performance characterization, or systems engineering. In fact, it intensified these needs. Large learned systems can be brittle under domain shift, opaque in their internal reasoning, expensive to evaluate exhaustively, and difficult to certify. They may perform impressively on broad benchmarks but fail in safety-critical edge cases. Their training data, deployment contexts, and human interfaces become part of the system.

The 2025 Stanford AI Index characterizes the scale and societal breadth of the current AI ecosystem, while NIST’s AI Risk Management Framework treats AI risk as a lifecycle and socio-technical problem[31, 29]. The EU AI Act, which entered into force in 2024 and becomes broadly applicable in stages, reflects the same shift from algorithmic performance to systems, risk, documentation, and governance[30]. CVPR 2025’s technical program highlighted areas such as 3D

computer vision, embodied computer vision, multimodal learning, and vision-language reasoning, which are all system-level and context-rich directions[32].

9.3 Foundation Models as a Return of Old Questions

Foundation models reintroduce several older themes.

- **Symbol grounding:** Vision-language models connect pixels to language but do not fully solve meaning, reference, or action.
- **Pattern synthesis:** Generative models revive analysis-by-synthesis and pattern theory questions.
- **World modeling:** Video and embodied models revive geometry, physics, dynamics, and control.
- **Systems engineering:** Deployment requires validation, monitoring, adaptation, safety constraints, and human oversight.
- **Philosophy of AI:** The question is no longer only “Can machines classify?” but “What kinds of cognitive architectures, agencies, and social roles should we build?”

Thus, the deep learning era should not be read as the end of the older traditions. It is a recombination of them under new computational conditions.

Chapter 10

Academic Integration: Goethe University and the 2011-2016 Transition

10.1 From Industrial Systems to Academic Synthesis

The 2011 transition to Goethe University Frankfurt marks a shift from building systems in industry to building frameworks, curricula, and transdisciplinary communities. The 2025 article describes this transition as a move from industrial foundations at Siemens to academic development at Goethe, including the Bernstein Focus in Neurotechnology from 2011 to 2016[8]. This period is crucial because it reframed the research program in relation to brain science, cognition, computation, and education.

The central question became: how can the lessons of systems engineering for vision be generalized into systems engineering for intelligence?

10.2 Cognition, Computation, Complexity

The 2025 curriculum article situates the broader educational transformation within a strategic profile of Computation, Cognition, and Complexity[8]. This triad is a natural culmination of the journey.

Computation provides formal languages, algorithms, architectures, and executable systems. Cognition provides questions about perception, attention, learning, reasoning, emotion, action, and meaning. Complexity provides the recognition that intelligent systems operate in dynamic, multi-scale, uncertain, socio-technical worlds.

10.3 From Vision Systems to Cognitive Architectures

The reflexive/reflective architecture from the 2008 deck becomes a bridge to cognitive systems. Fast components provide real-time responses. Slower components perform detailed estimation and

reasoning. Feedback updates world knowledge. Control mechanisms adjust sensing and processing parameters based on context. This is no longer just a vision pipeline; it is a cognitive architecture for task-oriented perception and action.

10.4 Perceptual Control Theory and Systems Engineering

Perceptual Control Theory gives a philosophical and psychological foundation for this shift. If behavior is the control of perception, then intelligent systems must be designed not merely to output labels but to maintain task-relevant perceptual variables under uncertainty. A surveillance system controls whether a person of interest remains in view. An autonomous vehicle controls perceived lane position, obstacle clearance, trajectory safety, and destination progress. An educational AI system may help regulate attention, feedback, and learning progress.

This connects the design of AI systems to control theory, psychology, and human-centered design.

Chapter 11

Certifiable Autonomous Cognitive Systems

11.1 The Dori-Ramesh 2015 Proposal

The 2015 Dori-Ramesh proposal on certifiable autonomous cognitive systems marks a further step in the journey. It argues that autonomous systems, with driving as a case in point, must integrate sensor analytics, conceptual modeling, quantitative performance characterization, and cognitive decision making[6]. It proposes combining Object-Process Methodology (OPM) with computer vision systems engineering to create an end-to-end model-based framework.

This is important because certification requires more than accuracy. It requires traceable requirements, formalized models, component interfaces, verification, validation, uncertainty modeling, and stakeholder communication. OPM’s graphical and textual bimodality supports both formal modeling and human scrutiny[6].

11.2 The Cyber-Physical Gap

The proposal introduces a key concept: the gap between the actual physical world and the system’s internal representation or world image[6]. This cyber-physical gap is central to AI safety. A self-driving car can fail because its perception system misdetects an object, but it can also fail because its internal model of the situation is outdated, incomplete, overconfident, or inconsistent with the physical world.

The cyber-physical gap connects technical and philosophical questions:

- What does the system believe about the world?
- How is that belief grounded in sensor evidence?
- How uncertain is the belief?

- What actions are safe given that uncertainty?
- How can the system know that it does not know enough?
- How can such knowledge be certified?

This is the dissertation’s perturbation-model question scaled up to autonomous cognitive systems.

11.3 Certification as a Systems Problem

Certification is sometimes treated as a regulatory burden. In this journey, it appears as a natural continuation of performance characterization. If a system’s behavior depends on context, task, sensors, algorithms, and uncertainty, then certification must reason about all of these. It cannot be reduced to a single benchmark score.

The 2015 proposal’s top-down and bottom-up architecture expresses the necessary duality. Top-down modeling captures functions, requirements, stakeholders, and system structure. Bottom-up algorithmic modeling captures sensor analytics, perception modules, learned components, and performance distributions. The certifiable system lives at their intersection.

Chapter 12

Educational Synthesis: Systems Engineering for Intelligence

12.1 The 2025 Curriculum Article

The 2025 article, *Systems Engineering for Intelligence: A Transdisciplinary Educational Approach*, makes the educational synthesis explicit. It describes a curriculum centered on the mapping from Context, Tasks, and Performance requirements to Hardware/Software System designs, $(C, T, P) \rightarrow \text{HW/SW Sys}$ [8]. It aims to train holistic systems thinkers who can develop AI systems that are safe, explainable, and effective.

The article also presents a pragmatic definition of intelligence, drawing from Aaron Sloman: intelligence as productive laziness or lazy productiveness[8]. This definition is pedagogically powerful. It tells students that intelligence is not just doing more computation; it is knowing what computation is worth doing, when, and why.

12.2 The Four Perspectives

The educational framework teaches systems design through four perspectives: user, modeling, implementation, and validation[8]. These perspectives recapitulate the journey.

Table 12.1: Four perspectives in systems engineering for intelligence

Perspective	Question	Historical root
User perspective	What context, stakeholders, needs, and acceptance criteria define the system?	Industrial deployment, HMI, application-domain modeling.

Modeling perspective	What formal representations of world, task, sensor, uncertainty, and performance are needed?	Pattern theory, statistical modeling, OPM, cognitive architectures.
Implementation perspective	How are models translated into algorithms, software, hardware, interfaces, and deployed components?	Vision pipelines, automatic morphology, software engineering.
Validation perspective	How do we know the system satisfies requirements under relevant conditions?	Performance characterization, simulation, experimental design, certification.

12.3 AISEL, ALI, and Project-Based Learning

The 2025 article describes AISEL, the AI Systems Engineering Laboratory, and ALI, AI and Digital Technologies in Learning and Instruction, as project-based educational vehicles[8]. AISEL applies systems engineering to student projects such as smart classrooms, AI Sudoku, greenhouse yield prediction, learning dynamics in deep networks, and simulation for continual learning. ALI extends the framework to educational technology, generative AI, affective AI, and learning environments.

This is a notable expansion. The same systems principles used for vision systems are now applied to education, human-AI interaction, and socio-technical design. That expansion is not accidental. Once one sees AI as a system embedded in context, any application domain requires user modeling, task modeling, performance criteria, implementation, and validation.

Chapter 13

Philosophy of AI, Sloman, and Productive Laziness

13.1 Why Philosophy Belongs in a Technical Journey

It may seem odd to place philosophy of AI inside a monograph about computer vision systems engineering. But the philosophical questions are unavoidable. What is a representation? What is an object? What is an action? What counts as understanding? What kinds of architectures can support flexible intelligence? How should a system decide what not to compute?

Aaron Sloman’s work is especially relevant because it treats philosophy as design-space exploration. Rather than arguing only about whether machines can think, Sloman asks what kinds of minds are possible and what architectures could generate them[22]. This style of philosophy is close to engineering, but it remains philosophical because it asks about possibilities, mechanisms, meanings, and explanatory frameworks.

13.2 Productive Laziness

Productive laziness is an excellent bridge between philosophy and systems engineering. A system with finite resources cannot evaluate every possible interpretation, every sensor action, every algorithm, and every parameter setting. It must use priors, context, abstractions, expectations, and heuristics. But these shortcuts must be principled, not merely ad hoc.

In the dissertation, productive laziness appears as parameter selection based on performance models. In Greiffenhagen’s system, it appears as uncertainty-aware zooming rather than blind high-resolution sensing everywhere. In the DARPA summary, it appears as active sensing and performance prediction. In the 2008 CVPR deck, it appears as reflexive hypothesis generation followed by reflective reasoning. In the 2025 curriculum, it appears as a definition of intelligence for education[8].

13.3 Architectures, Not Single Methods

Sloman also helps guard against reductionism. AI has repeatedly been tempted by single-method narratives: logic will solve intelligence; neural networks will solve intelligence; probabilistic models will solve intelligence; language models will solve intelligence. The journey reconstructed here suggests a different view: intelligent systems require architectures of interacting mechanisms.

Those mechanisms include sensing, filtering, feature extraction, learned representation, symbolic abstraction, probabilistic inference, control, memory, planning, self-monitoring, human interaction, and validation. No single mechanism is enough.

Chapter 14

Lessons for Students

14.1 Lesson 1: Learn to Ask What the System Is For

The first lesson is to begin with context, tasks, and performance. A system cannot be designed in the abstract. A detector for a medical system, an autonomous vehicle, a classroom tool, and a security camera may all use “AI,” but they inhabit different worlds, stakeholders, errors, consequences, and acceptance criteria.

14.2 Lesson 2: Treat Algorithms as Components in Worlds

An algorithm is not merely a function from input to output. In a deployed system, it is a component in a world. It has assumptions, parameters, failure modes, interfaces, timing constraints, data dependencies, and social consequences. Students should learn to ask how the algorithm behaves under perturbation, not only how it performs on a leaderboard.

14.3 Lesson 3: Measure, Model, and Validate

Performance characterization is a form of disciplined humility. It asks the system designer to state what is known, what is assumed, what varies, and how uncertainty propagates. This is as important for deep networks as for edge detectors.

14.4 Lesson 4: Use Theory and Data Together

The journey resists the false opposition between model-based and data-driven methods. Mathematical models provide structure, interpretability, and extrapolation. Data provides empirical grounding and adaptation. The best systems often combine both.

14.5 Lesson 5: Build for Reuse and Adaptation

Industrial systems teach that robust design is not only about solving today's problem. It is about making the system extensible, diagnosable, reusable, and adaptable to new conditions.

14.6 Lesson 6: Study Philosophy and Psychology

Students building AI systems should study perception, cognition, control, and philosophy of mind. These disciplines provide questions that prevent narrow engineering. They also provide conceptual tools for human-AI interaction, agency, explanation, trust, and meaning.

14.7 Lesson 7: Design for Human Flourishing

The final lesson is ethical and educational. AI systems should enhance human capabilities rather than diminish them. They should make decisions understandable, preserve agency, support learning, and operate within accountable institutions. This is why systems engineering for intelligence is not merely technical training; it is preparation for responsible technological citizenship[8].

Chapter 15

A Webpage Blueprint for the CCC Site

15.1 Purpose of the Webpage

The envisioned webpage should not be merely a CV. It should be a guided intellectual journey. It should allow a young student to enter at multiple levels: biography, technical ideas, history of computer vision, systems engineering, teaching, philosophy, and future questions.

15.2 Suggested Navigation

A possible structure is:

1. **Welcome: Counting with Uncertainty.** A short reflective introduction.
2. **The Three Strata.** University research, industrial scaling, academic integration.
3. **Mentors and Lineages.** Robert Haralick, Thomas Binford, Hyonam Joo, Aaron Sloman, and other influences.
4. **Core Ideas.** Performance characterization, application priors, reflexive/reflective vision, context-task-performance mapping.
5. **Documents and Artifacts.** CVPR08 deck, dissertations, papers, 2018 article, 2025 curriculum article.
6. **For Students.** Reading paths, project ideas, questions to ask, advice on transdisciplinary work.
7. **Future Directions.** Safe human-AI co-evolution, certifiable cognitive systems, education, governance.

15.3 Reading Paths for Students

The page could offer multiple reading paths.

Table 15.1: Suggested webpage reading paths

Student interest	Suggested path
Computer vision algorithms	Haralick texture, mathematical morphology, Ramesh dissertation, CVPR08 history slides.
AI and symbolic reasoning	Dartmouth proposal, Joo dissertation, Sloman, rule-based reasoning in video surveillance.
Robotics and autonomy	Active vision, Greiffenhagen dual-camera system, Dori-Ramesh certifiable autonomous driving.
AI safety and governance	Performance characterization, cyber-physical gap, NIST AI RMF, EU AI Act, validation perspective.
Education and transdisciplinary computing	2025 curriculum article, AISEL, ALI, CCC profile, project-based learning.
Philosophy and psychology	Marr, Gibson, Powers, Sloman, reflexive/reflective architecture.

15.4 Tone

The tone should be humble, grateful, and intellectually generous. The webpage should communicate that the journey was blessed not because it was easy, but because it brought together mentors, students, colleagues, industrial problems, academic freedom, and the opportunity to keep learning.

Chapter 16

Conclusion: From Computer Vision to Systems Engineering for Intelligence

The journey reconstructed in this monograph begins with computer vision but does not end there. It begins with images, features, and pattern recognition. It passes through mathematical morphology, symbolic AI, theorem proving, statistical perturbation modeling, performance characterization, real-time surveillance, active sensing, and industrial deployment. It then becomes a broader academic program: systems engineering for intelligence, rooted in computation, cognition, and complexity.

The unifying thread is the attempt to build systems that know something about their own limits. A system should not merely output an answer. It should be designed from context, task, and performance requirements. It should model its uncertainty. It should adapt its sensing and computation. It should support explanation and validation. It should be teachable, extensible, and certifiable when the application demands it. It should be designed with humans, not merely for humans.

From the 1960s and 1970s to 2026, the central questions have changed form but not substance. How do signals become patterns? How do patterns become objects? How do objects become situations? How do situations guide action? How do systems know when they are wrong? How can we teach students to build such systems responsibly?

A blessed journey is not only a sequence of achievements. It is a sequence of invitations: to learn from mentors, to serve real-world needs, to cross disciplinary boundaries, to guide students, and to keep asking what intelligence should mean in a world shared by humans and machines.

Appendix A

Timeline of People, Ideas, and Artifacts

Table A.1: A compact timeline

Period	Field context	Journey/artifact context
1940s-1950s	Information theory, cybernetics, early neural models, early AI ambitions.	Conceptual ancestors: Shannon, Wiener, McCulloch-Pitts, Rosenblatt.
1960s-1970s	Statistical pattern recognition, symbolic AI, theorem proving, image processing, mathematical morphology, pattern theory, psychology of vision.	Foundations later synthesized in the CVPR08 history slides.
1980s	Modular vision systems: feature extraction, grouping, matching, model-based vision, geometry.	Graduate education context begins; Haralick lineage.
1991	Symbolic AI and morphology meet in automatic procedure synthesis.	Hyonam Joo dissertation, <i>Automatic Morphology</i> .
1995	Systems engineering enters image-understanding algorithms.	Ramesh dissertation, <i>Performance Characterization of Image Understanding Algorithms</i> .
1995-2010	Commercialization of computer vision, surveillance, automotive, healthcare, multisensor systems.	Siemens Corporate Research; real-time vision and modeling.

2001	Real-time active camera systems and uncertainty-aware control.	Greiffenhagen dissertation.
2008	Computer vision field anticipating multimodal, cognitive, active, contextual futures.	CVPR panel presentation on future directions.
2011-2016	Cognitive systems, neurotechnology, transdisciplinary computing.	Goethe University Frankfurt transition; Bernstein Focus in Neurotechnology.
2013	Performance modeling for systems of systems.	DARPA performance-modeling summary.
2015	Model-based certification of autonomous cognitive systems.	Dori-Ramesh proposal using OPM and driving as case.
2018	Public-facing AI reliability and uncertainty.	Frankfurt article, <i>Kuenstliche Intelligenz: Wie verlaesslich ist sie?</i>
2025	Systems engineering for intelligence as education.	Curriculum article describing $(C, T, P) \rightarrow$ HW/SW Sys, AISEL, ALI, CCC.
2026	Foundation models, multimodal AI, embodied AI, AI risk management and regulation.	Retrospective context for revamping webpage and mentoring students.

Appendix B

Glossary

Application domain model

A representation of the relevant conditions, priors, constraints, and variability of the world in which a system operates.

Cyber-physical gap

The discrepancy between the actual physical world and the system's internal representation or world image.

Performance characterization

The process of modeling how input perturbations, algorithm parameters, and context affect output perturbations and task performance.

Productive laziness

A Sloman-inspired notion of intelligence as solving problems efficiently by knowing what can be ignored, abstracted, or deferred.

Reflexive vision

Fast, real-time hypothesis generation, often using priors and efficient transforms.

Reflective reasoning

Slower, more detailed reasoning, estimation, belief revision, and validation.

Systems engineering for intelligence

A transdisciplinary methodology for mapping context, tasks, and performance requirements to hardware/software systems that are safe, explainable, and effective.

White-box performance modeling

Analytical modeling of component and system behavior through explicit deterministic and stochastic representations.

Black-box evaluation

Empirical end-to-end evaluation without explicit modeling of component internals.

Appendix C

Annotated Source Map

Table C.1: Primary sources used in this draft

Source	Role in the monograph
CVPR08 future-directions deck[1]	Provides the historical hinge: early vision history, systems engineering view, reflexive/reflective architecture, and future topics.
Ramesh dissertation[2]	Provides the core methodology of performance characterization, perturbation models, application-domain priors, and algorithm tuning.
Hyonam Joo dissertation[3]	Provides the symbolic-AI and automatic-programming complement through predicate-logic morphology.
Greiffenhagen dissertation[4]	Provides the industrial scaling example: real-time dual-camera active sensing with uncertainty-aware control.
DARPA performance-modeling summary[5]	Provides the mature formulation of white-box/black-box performance modeling, active vision, diagnostics, and systems of systems.
Dori-Ramesh proposal[6]	Provides the certifiable-autonomy and OPM-based systems-engineering layer.
2018 Frankfurt article[7]	Provides the public-facing formulation of AI reliability and “counting with uncertainties.”
2025 curriculum article[8]	Provides the educational synthesis: $(C, T, P) \rightarrow$ HW/SW Sys, productive laziness, AISEL, ALI, CCC, and holistic systems thinkers.

Appendix D

Addendum: Publication Lineage, Citation Footprint, and the Research Program Behind the Journey

This addendum reads the publication record not as a bibliography to be counted but as a lineage to be interpreted. The main online anchors are the Google Scholar profile listed in the January 2026 short CV and the dblp profile for Visvanathan Ramesh. The short CV reports the Google Scholar profile as having 22,586 citations, h-index 50, i10-index 114, and 45 awarded US patents as of January 28, 2026[33]. The dblp profile gives the curated computer-science publication lineage, ORCID 0000-0002-8842-905X, affiliations spanning Goethe University Frankfurt, FIAS, and Siemens Corporate Research, and records across the 1990s, 2000s, 2010s, and 2020s[34].

D.1 Why a Publication Lineage Matters

A publication list is often read in two shallow ways: as a productivity metric or as a set of isolated technical contributions. For this monograph, neither reading is sufficient. The papers matter because they reveal a long-running research program: how to design, analyze, validate, and deploy intelligent vision systems under uncertainty. Across more than three decades, the titles shift from random perturbation models, tuning parameters, boundary extraction, and morphology to mean-shift tracking, video monitoring, multimodal representations, object recognition, active sensing, image grammars, simulation, continual learning, transparent deep models, and educational AI. The continuity is not a single algorithm. It is the question of how perception becomes reliable enough for action.

This addendum therefore interprets the record as a set of intertwined threads. Some papers are heavily cited because they became general-purpose tools for the community. Others are less visible in citation counts but are central to the deeper intellectual arc: performance characterization, formal representation of operators, domain modeling, system validation, and the movement from

computer vision to systems engineering for intelligence.

D.2 Two Bibliographic Lenses: Google Scholar and dblp

Google Scholar and dblp serve different purposes. Google Scholar is broad and citation-oriented. It captures the impact footprint across computer vision, machine learning, engineering, and adjacent literatures. The January 2026 CV records a Google Scholar citation count above 22,000, with h-index 50 and i10-index 114[33]. Search-engine snippets visible during the preparation of this addendum also listed the profile as “Cited by” slightly above 22,500, which is consistent with the January 2026 CV while reflecting ordinary indexing differences over time[35]. These numbers should be read as approximate indicators of influence, not as the substance of the journey.

Dblp, by contrast, is a curated computer-science bibliography. It is especially useful for reconstructing lineage because it arranges publications by year, venue, coauthors, and type. The dblp page lists early University of Washington work from the 1990s, the Siemens-era cluster of papers in the 2000s, the cognitive-systems and simulation work after the move to Frankfurt, and the recent deep-learning, continual-learning, hybrid-network, and education-technology papers of the 2020s[34]. In a webpage redesign, the two lenses should be used together: Google Scholar for impact, dblp for intellectual genealogy.

D.3 A Lineage in Eight Research Threads

The following table is deliberately interpretive. It does not include every paper. Instead, it groups representative publications into the intellectual themes that make the journey legible to a student.

Table D.1: Representative paper lineage by research thread

Thread	Representative papers and venues	What the thread contributed to the journey
Random perturbation and performance characterization	“Random perturbation models and performance characterization in computer vision” (CVPR 1992); “Automatic selection of tuning parameters for feature extraction sequences” (CVPR 1994); “Random perturbation models for boundary extraction sequence” (Machine Vision and Applications, 1997)[34].	Established the central methodological question: how do input perturbations, algorithm parameters, and domain priors shape output errors and task performance? This is the seed of the later systems-engineering framework.

Thread	Representative papers and venues	What the thread contributed to the journey
Formal image operators, morphology, and algorithm generation	“MUSER: A prototype musical score recognition system using mathematical morphology” (Machine Vision and Applications, 1993); Hyonam Joo’s <i>Automatic Morphology</i> ; later “Semi-Automatic Probabilistic Morphological Detection” (ICASSP 2005) and “Predicate logic based image grammars for complex pattern recognition” (CVPR Workshops 2009)[3, 34].	Kept alive the symbolic-AI and automatic-programming dream inside image processing: task knowledge, operator knowledge, and formal representations can help synthesize or structure visual procedures.
Shape, motion, calibration, and physical image formation	“Error Characterization of the Factorization Approach to Shape and Motion Recovery” (Workshop on Vision Algorithms, 1999; CVIU, 2001); “Statistical Calibration of the CCD Imaging Process” (ICCV 2001); “A Class of Photometric Invariants: Separating Material from Shape and Illumination” (ICCV 2003)[34].	Connected performance modeling to geometry, sensor physics, illumination, and calibration. This thread is crucial for bridging abstract statistical models with the physical conditions under which systems actually operate.
Mean shift, kernel tracking, and statistical real-time tracking	“Real-Time Tracking of Non-Rigid Objects Using Mean Shift” (CVPR 2000); “Mean Shift and Optimal Prediction for Efficient Object Tracking” (ICIP 2000); “The Variable Bandwidth Mean Shift and Data-Driven Scale Selection” (ICCV 2001); “Kernel-Based Object Tracking” (IEEE TPAMI, 2003)[34].	Created a highly influential real-time tracking lineage. Its impact came from the combination of nonparametric statistics, efficient optimization, practical robustness, and immediate relevance to video understanding.

Thread	Representative papers and venues	What the thread contributed to the journey
Real-time video monitoring and active sensing systems	“Statistical Modeling and Performance Characterization of a Real-Time Dual Camera Surveillance System” (CVPR 2000); “The Systematic Design and Analysis Cycle of a Vision System” (CVPR 2001); “Design, analysis, and engineering of video monitoring systems” (Proceedings of the IEEE, 2001); “A MRF-Based Approach for Real-Time Subway Monitoring” (CVPR 2001)[34, 4].	Scaled algorithmic ideas into deployed architectures with requirements, active camera control, uncertainty-aware zooming, reuse of analysis, and performance limits. This is the industrial systems-engineering turn.
Object recognition, small-sample learning, and robustness	“Bayesian Color Constancy for Outdoor Object Recognition” (CVPR 2001); “On the Small Sample Performance of Boosted Classifiers” (CVPR 2005); “Sudden illumination change detection using order consistency” (Image and Vision Computing, 2004); “Learn to Track Edges” (ICCV 2007)[34].	Addressed robustness under limited data, lighting changes, and dynamic scenes. These papers anticipate later concerns in domain shift, open-world learning, uncertainty, and safety-critical evaluation.
Simulation, cognitive vision, and validation of vision systems	“Simulations for Validation of Vision Systems” (2015); “Model-driven Simulations for Deep Convolutional Neural Networks” (2016); “Adversarially Tuned Scene Generation” (CVPR 2017); automotive anomaly-detection work in 2017[34].	Reframed simulation as a controlled experimental instrument for performance characterization, systematic evaluation, and model-driven design. This thread directly supports certification and safe AI by design.

Thread	Representative papers and venues	What the thread contributed to the journey
Continual learning, transparent deep models, hybrid neural systems, and education	“Meta-Learning Convolutional Neural Architectures for Multi-Target Concrete Defect Classification” (CVPR 2019); “A Wholistic View of Continual Learning with Deep Neural Networks” (Neural Networks, 2023); “Neural Architecture Search of Deep Priors” (CVPR Workshops 2021); “When Deep Classifiers Agree” (ECCV 2022); 2024–2025 work on decomposed encoders, hybrid networks, uncertainty-aware networks, distribution shifts, stable-plastic neural networks, and AI/digital-technology competence in educational processes[34].	Shows the Frankfurt-era integration: deep learning is not treated as a replacement for systems thinking but as a new class of components requiring design, transparency, continual adaptation, evaluation, and human-centered educational framing.

D.4 The Citation Peak and the Hidden Backbone

The best-known visible peak in the publication record is the mean-shift and kernel-tracking lineage. The CVPR 2000 paper on real-time tracking of non-rigid objects and the TPAMI 2003 paper on kernel-based object tracking became durable references in computer vision. The public search results available during this addendum preparation listed thousands of citations for these works, with the TPAMI paper above seven thousand citations on an ACM-indexed page and the CVPR paper also appearing with several thousand citations in PDF/search snippets[35, 34]. This peak is real and important. It helped make the work visible to generations of researchers who needed fast, robust, nonparametric tracking.

But a citation peak can hide the backbone of a research life. The backbone here is not only tracking. It is the long-running insistence that vision algorithms are not isolated functions. They are components in larger systems, operating under noisy data, changing domains, limited computation, sensor constraints, and user-specified requirements. The 1992–1997 performance-characterization papers, the 1995 dissertation, the Greiffenhagen system, the DARPA 2013 performance-modeling summary, and the 2015 certifiable-autonomy proposal are less easily compressed into a single algorithmic keyword, but they form the methodological spine of the journey[2, 4, 5, 6].

This distinction is pedagogically important. Students often learn to chase the paper that is

most cited. A healthier lesson is to ask: what problem did the researcher refuse to abandon? In this journey, the persistent problem is how to make machine perception accountable to context, uncertainty, performance, and action.

D.5 Coauthor Communities as Intellectual Habitats

The dblp coauthor structure is another clue to the journey. Early coauthors such as Robert M. Haralick, Linda Shapiro, Xining Zhang, and collaborators in the DARPA Image Understanding Environment link the work to statistical pattern recognition, image understanding, mathematical morphology, and performance characterization. The Siemens-era coauthor community - including Dorin Comaniciu, Peter Meer, Michael Greiffenhagen, Yanghai Tsin, Nikos Paragios, Terrance Boult, Binglong Xie, Weiliang Li, and others - reflects the move to real-time systems, industrial scale, active sensing, surveillance, object recognition, and robust tracking. The Frankfurt-era coauthors - including Constantin Rothkopf, V. S. R. Veeravasaru, Martin Mundt, Tobias Weis, Iuliia Pliushch, Achref Jaziri, and colleagues in education and cognitive science - reflect a shift toward cognitive architectures, simulation, continual learning, transparent deep models, and AI systems engineering in education[34, 33].

This coauthor map should be treated as more than a network diagram. It is a map of intellectual habitats. Each habitat carried different constraints: graduate research, DARPA evaluation culture, corporate R&D, customer-facing product pressure, neurotechnology, educational design, and AI governance. The same person moved through these habitats with a recurring methodological style: define the task, model the context, characterize uncertainty, design the system, validate the behavior, and revise the architecture.

D.6 Mentoring Lineage and the Expansion from Papers to People

The January 2026 CV makes the mentoring lineage explicit. At Goethe University, the supervised Ph.D. topics include simulations for cognitive vision, continual learning in an open world, cognitive vision architectures, and transparent deep learning models. Earlier industrially advised Ph.D. topics include real-time dual-camera surveillance, low-level vision performance characterization, boosting in computer vision, face detection and recognition, watching humans and activities, kernel correlation, object-based video processing with depth, and spatial invariance models[33]. This is the human continuation of the paper lineage.

The webpage redesign should therefore not simply list students. It should show that each student represents a branch of a systems question:

- How can simulations support systematic evaluation of vision systems?
- How can learning continue in an open world without catastrophic forgetting?
- How can cognitive vision architectures combine fast perception and reflective control?

- How can transparent deep models be designed rather than merely explained after the fact?
- How can real-time video systems offer performance guarantees in real deployments?

This converts a CV into a mentorship map. For young students, that is far more illuminating than a chronological list.

D.7 From Publications to a Webpage Architecture

The addendum suggests a concrete structure for the revamped webpage. Rather than organizing the site only by positions or publication years, the homepage could present the journey through four navigable layers.

Table D.2: Publication-lineage-driven webpage architecture

Webpage section	Core material	Student-facing message
Origins: learning to characterize uncertainty	UW dissertation, Haralick mentorship, perturbation models, automatic tuning, image understanding environment.	Scientific rigor begins by asking what can go wrong and how to measure it.
Scaling: making vision work in the world	Siemens real-time vision, mean-shift tracking, dual-camera surveillance, video monitoring, patents, products, customer-facing systems.	Real-world AI requires architecture, teams, constraints, validation, and responsibility.
Integration: cognition, simulation, and systems science	Bernstein Focus, cognitive vision, model-driven simulation, continual learning, hybrid architectures.	AI systems are not just models; they are situated cognitive-technical systems.
Education: preparing holistic systems thinkers	AISEL, ALI, $(C, T, P) \rightarrow$ HW/SW Sys, CCC, philosophy of computation, productive laziness.	The next generation must combine computation, cognition, complexity, ethics, and design.

D.8 A Suggested Narrative for the Publication Page

The publication page could begin with a short narrative rather than immediately displaying a complete list:

My publications are best read as a journey from uncertainty in image understanding algorithms to systems engineering for intelligence. The early work asked how random perturbations propagate through vision modules. The Siemens years asked how such analysis could scale to real-time systems, tracking, surveillance, industrial imaging, healthcare, and active sensing. The Frankfurt years ask how model-based design, simulation, continual learning, transparent deep architectures, and educational systems can prepare us for safe human-AI co-evolution. Citation counts highlight some peaks, especially the mean-shift and kernel-tracking papers, but the deeper theme is the design of intelligent systems whose behavior can be specified, analyzed, validated, and trusted.

The complete publication list can then be linked through Google Scholar and dblp, while selected papers are presented as “lineage anchors” grouped under the threads in Table D.1. This would let a young reader see not only what was published, but why the papers belong together.

D.9 Retrospective Synthesis

The paper lineage strengthens the central thesis of the monograph. The path is not from old computer vision to new AI in a simple replacement story. It is a path from measured visual evidence to systems that must act in the world. The early mathematical and statistical work on perturbations made visible the need for performance characterization. The mean-shift lineage showed how statistical insight can become a widely used real-time tool. The video monitoring papers showed how algorithms become systems. The simulation and cognitive-vision papers showed how evaluation itself becomes an object of design. The recent continual-learning, hybrid-network, and educational-technology papers show that deep learning must be embedded in systems thinking, not treated as an escape from it.

For a student visiting the future webpage, the lesson should be simple and generous: a blessed journey in AI is not built by chasing fashion. It is built by returning, again and again, to questions that matter - uncertainty, reliability, context, action, explanation, validation, and human flourishing.

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